R13

Code No: 126AQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, December - 2017
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Used paper 13-12-2017

PART - A

(25 Marks)

1.a)	Define Non Repudiation.	[2]
b)	Write a short notes on steganography.	[3]
c)	Define linear cryptanalysis.	[2]
d)	Discuss about Electronic code book mode?	[3]
e)	Define Message Authentication Code.	[2]
f)	Illustrate about biometric authentication.	[3]
g)	What is IP Security?	[2]
h)	Discuss about the concept of combining security associations.	[3]
i)	What is Firewall?	[2]
j)	Write short notes on virtual elections.	[3]

PART - B

(50 Marks)

2. Compare and Contrast between Symmetric and Asymmetric key cryptography. [10]

OR

3. Give an example to explain the concept of transposition ciphers in detail. [10]

4. With a neat diagram explain how encryption and decryption are done using Blowfish algorithm? [10]

OR

5. Given two prime numbers p=5 and q=11, and encryption key e=7 derive the decryption key d. Let the message be x=24. Perform the encryption and decryption using R.S.A algorithm. [10]

6. Give a neat sketch to explain the concept of Secured Hash Algorithm (SHA). [10]

OR

7. Client machine C wants to communicate with server S. Explain how it can be achieved through Kerberos protocol? [10]

8. How the messages are generated and transmitted in pretty good privacy (PGP) protocol? Explain with clear diagrams. [10]

OR

9. Draw the IP security authentication header and explain the functions of each field. [10]

10. Explain the steps involved in performing Secure Inter-branch Payment Transactions. [10]

OR

11. List the characteristics of a good firewall implementation? How is circuit gateway different from application gateway? [10]

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